

## THERMACART · CALCULATION

# ThermaCart sizing calculations

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## DOCUMENT CONTROL

|             |                                         |         |                |
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| Project TRL | <b>TRL 3</b> Proof of concept           |         |                |
| Repository  | github.com/BoujeeEnjinia1701/thermacart |         |                |
| Source      | docs/04-calcs/01-sizing.md @ 390f06f    |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                           |
|---------|------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); dark finish as the design case, paint mass and cost added, R1, R5 and R14 restated, results table updated           |
| 0.1     | 2026-09-25 | Amish Chadha | First issue for TRL 3 (geometry and mass, storage per grade, pressure and wall stress, 2 mm wall option, hold times, recharge, handle temperature, keying, cost) |

# ThermaCart sizing calculations

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On paper the TC-L cartridge meets twelve of its eighteen requirements, has one at risk, misses two and leaves three that only tests can settle. This version applies the recommendations Amish accepted on 2026-09-25 (TCT-DDR-002): the shell carries a matte black finish (O8), so the dark-finish results are now the design values; the C25 band of R1 is 22 to 27 °C (O9); R5 is 38 % for the stock-tube build (O4); and R14 includes a labelled handling rule for H70 cartridges from an oven (O7). The two misses are the cold-chain hold time (R10: 2.4 h against 8 h, because the cartridge cannot pull heat out of the cooler air fast enough, although it stores enough energy for 9.7 h) and the parts cost (R16: \$76 per cartridge against \$50, which also puts one cartridge and one frame \$12 over the \$80 budget). The one at risk is recharge on a 60 W pad (R12). With the verified Rubitherm datasheet values, the C5 grade now meets R3 (65 Wh), which the TRL 2 estimate did not; C25 meets it by a thin margin. The 2 mm wall evaluated under decision D5 is rejected: it adds only 3.5 % volume and its walls would yield under the vacuum that forms when a cartridge is frozen. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C3], is the line of that script's output that carries it.

**Safety:** These are first-principles estimates for a paper proof of concept. They do not show that a cartridge is leak-tight, drop-safe or fit for food contact, and they are not a substitute for tests. ThermaCart is not qualified for vaccines or medicines. See TCT-PRC-001, Safety.

## ■ Scope and method

The note checks every requirement in TCT-REQ-001 v0.4 against the design in TCT-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, its derived dimensions and its `build123d` solids, so the inner volume, part volumes, areas and key positions used here are those in the STEP files and in drawing TCT-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

## ■ Assumptions

*Table 1. Main assumptions.*

| Area                  | Assumption                                                                                                                                                                                                                                                                                                                                                                                                                   | Basis                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| PCM data              | RT 5 HC: 250 kJ/kg from -2 to 13 °C, melt 5 to 6 °C, 0.85 kg/L solid, 0.76 kg/L liquid at 20 °C, maximum operating temperature 45 °C. RT 25 HC: 230 kJ/kg from 16 to 31 °C, melt 22 to 26 °C, 0.88 and 0.77 kg/L (liquid at 40 °C), maximum 65 °C. RT 70 HC: 260 kJ/kg from 62 to 77 °C, melt 69 to 71 °C, about 0.9 and 0.8 kg/L (liquid at 80 °C), maximum 90 °C. All: cp 2 kJ/(kg·K), k 0.2 W/(m·K), capacity $\pm 7.5\%$ | Rubitherm datasheets for <a href="#">RT 5 HC</a> , <a href="#">RT 25 HC</a> and <a href="#">RT 70 HC</a> , read 2026-09-25 |
| Usable storage        | Latent heat = capacity at the low side of the tolerance minus 2 kJ/(kg·K) x 15 K sensible; usable = latent plus sensible over the $\pm 3$ K band of R3; latent spread evenly over the melt range                                                                                                                                                                                                                             | Conservative reading of the datasheets                                                                                     |
| Fill                  | 90 % of the inner volume, filled as liquid at the grade's maximum operating temperature (45, 65 and 90 °C), then sealed with air at 1 atm; liquid expansion 0.001 /K, solid 0.0003 /K                                                                                                                                                                                                                                        | Expansion coefficients are typical paraffin values, to confirm                                                             |
| Aluminium             | 6063-T52 tube, minimum yield 110 MPa, design factor 1.5 (allowable 73 MPa); E 69 GPa; 2.70 kg/L, 0.9 kJ/(kg·K)                                                                                                                                                                                                                                                                                                               | ASTM B221 minimum                                                                                                          |
| Air side              | Natural convection $h = 1.32 (\Delta T/0.15 \text{ m})^{0.25} \text{ W}/(\text{m}^2\cdot\text{K})$ plus radiation; emissivity 0.90 for the decided matte black finish (TCT-DDR-002, O8), 0.10 for mill-finish aluminium as a comparison; 85 % of the outer area exposed                                                                                                                                                      | Handbook correlations                                                                                                      |
| Cold box (R10)        | 25 L cooler, 0.5 W/K, 32 °C ambient; 5 kg of produce precooled to 5 °C (3.9 kJ/(kg·K)) plus 2 kJ/K of air and liner; cartridges conditioned to 2 °C; contents and cooler air lumped, no direct contact between produce and cartridge                                                                                                                                                                                         | As in TCT-REQ-001; contact would help and is not credited                                                                  |
| Hot box (R11)         | GN carrier, 0.25 W/K, 25 °C ambient; 4 kg of food at 75 °C (3.5 kJ/(kg·K)) plus 2 kJ/K of liner; H70 charged to 85 °C                                                                                                                                                                                                                                                                                                        | As in TCT-REQ-001                                                                                                          |
| Charging (R12)        | Still-air freezer at -18 °C, refrigerator at 2 or 4 °C, fan oven at 85 °C ( $h = 15 \text{ W}/(\text{m}^2\cdot\text{K})$ plus radiation), 60 W pad with 80 % of its output entering the cartridge                                                                                                                                                                                                                            | Screening values                                                                                                           |
| Burn thresholds (R14) | 10 s contact: about 55 °C for bare metal and about 70 °C for plastics and rubber                                                                                                                                                                                                                                                                                                                                             | Commonly quoted from ISO 13732-1; not checked against the standard (paywalled)                                             |
| Cost                  | Indicative one-off USD prices in <code>bom/bom.csv</code> , not quotes; PCM at \$10/kg                                                                                                                                                                                                                                                                                                                                       | To be replaced with quotes before any build                                                                                |

## ■ A. Geometry and mass (R2, R4, R5)

- **Envelope.** With the bail stowed the cartridge is 323.4 x 152.4 x 63.5 mm, inside the GN 1/3 slot (325 x 176 mm, 65 mm deep) with margins of 1.6, 23.6 and 1.5 mm [A1]. The TRL 2 fixed handle left only about 14 mm between grip and end cap, too little for fingers; the folding bail stows inside the envelope and gives a 39 mm finger gap when raised [A3].
- **Inner volume.** 146.05 x 44.45 x 254.9 mm, or 1.655 L [A2], slightly more than the 1.62 L of TRL 2, because a thinner 3.175 mm (1/8 in) flange let the tube grow from 270 to 280 mm.
- **Mass.** The model's aluminium parts come to 1.665 kg, and seals, screws, plug, grip, label, adhesive and about 15 g of primer and paint add 0.130 kg, so the empty cartridge is 1.80 kg [A4, A5] against about 1.5 kg at TRL 2. The caps are the reason: a garage-built cap is a solid stack of flange, gland spacer and plug (about 100 cm<sup>3</sup> each) rather than the milled, pocketed cap assumed at TRL 2.

- **Filled.** C5 2.90 kg, C25 2.91 kg and H70 2.98 kg, with 1.104, 1.119 and 1.180 kg of PCM [A6]. R4 (3.0 kg) is met with a thin margin. The PCM fraction is 38.1 to 39.7 %, which meets the 38 % target set for the stock-tube build by TCT-DDR-002 (O4) with little margin; the 50 % target stays for a later extrusion.

## ■ B. Usable storage per grade (R1, R3)

R1 is met for all three grades. RT 25 HC melts over 22 to 26 °C, which lies inside the C25 band of 22 to 27 °C set by TCT-DDR-002 (O9); the earlier 24 to 27 °C band would not have been met.

Table 2. Storage within  $\pm 3$  K of the melt point [B1].

| Grade | PCM      | Latent (low side) | Usable in band | Per cartridge  | At nominal datasheet value |
|-------|----------|-------------------|----------------|----------------|----------------------------|
| C5    | 1.104 kg | 201.2 kJ/kg       | 213.2 kJ/kg    | <b>65.4 Wh</b> | 71.2 Wh                    |
| C25   | 1.119 kg | 182.8 kJ/kg       | 194.8 kJ/kg    | <b>60.5 Wh</b> | 65.9 Wh                    |
| H70   | 1.180 kg | 210.5 kJ/kg       | 222.5 kJ/kg    | <b>72.9 Wh</b> | 79.3 Wh                    |

- All three grades meet R3 (60 Wh). The TRL 2 figures (55, 61 and 67 Wh) used assumed usable heats of 180, 200 and 220 kJ/kg; the datasheets support more for C5 and H70 and less for C25.
- Energy density on filled mass is 22.6, 20.8 and 24.5 Wh/kg [B2].
- **2 mm wall (D5).** A metric 150 x 50 x 2 mm tube of the same length holds 1.712 L, only 3.5 % more, and would raise C5 to 67.7 Wh [B3]. Section C shows its walls would yield. It is not needed for R3 and is not adopted.

## ■ C. Pressure and wall stress (R8, D5)

Filling as liquid at the grade's maximum operating temperature fixes the gauge pressure at zero at that temperature. Everything colder makes a partial vacuum, because the PCM shrinks by about 12 % on freezing and the trapped air cools.

Table 3. Gauge pressure with 10 % ullage [C1].

| Grade | Sealed at | Solid at -25 °C           | 20 K above the limit |
|-------|-----------|---------------------------|----------------------|
| C5    | 45 °C     | -0.63 bar (ullage 20.8 %) | +0.29 bar at 65 °C   |
| C25   | 65 °C     | -0.68 bar (ullage 22.1 %) | +0.29 bar at 85 °C   |
| H70   | 90 °C     | -0.64 bar (ullage 18.5 %) | +0.29 bar at 110 °C  |

- **R8 is met:** 0.00 bar at the fill temperature by construction and +0.29 bar even 20 K beyond the datasheet limit, against 0.5 bar [C2].
- **Walls.** Treating the tube as a closed frame loaded on all four walls, the corner bending stress is 56 MPa at -0.68 bar, 83 MPa at full vacuum and 24 MPa at +0.29 bar, against an allowable of 73 MPa [C3]. The 3.175 mm wall carries the worst service vacuum with margin. Full vacuum could only arise in a cartridge sealed with no air; it would exceed the allowable stress but not the yield. The broad faces deflect about 0.8 mm inward at -0.68 bar [C5].
- **2 mm wall (D5).** The same vacuum gives 141 MPa, 1.28 times the minimum yield; its allowable pressure is only 0.35 bar [C4]. Rejected.

- **Caps.** At +0.29 bar each cap carries 190 N, 32 N per radial M5 screw [C6]. Under vacuum the flange bears on the tube end.
- **Drop (R6, indicative).** A molten cartridge dropped 1 m end-on and stopping in 4 mm decelerates at about 250 g; if the gas ullage did not cushion the liquid, the surge could reach 5.0 bar on the lower cap, 541 N per screw [C7]. The screws have ample capacity (M5 A2-70 proof load about 6,400 N), but that surge is ten times the service pressure on the walls and the O-ring. R6 needs a drop test at TRL 4.

## ■ D. Hold times (R10, R11)

The outer area is 0.1138 m<sup>2</sup> of tube, 0.0503 m<sup>2</sup> of fins and 0.0155 m<sup>2</sup> of end faces, 0.1526 m<sup>2</sup> effective [D1].

**R10 is not met.** Two C5 cartridges store 131 Wh within the band, enough for 9.7 h at the 13.5 W leak [D2], so energy is not the limit. The limit is heat transfer from the cooler air into the cartridges. With the decided matte black finish (emissivity 0.9), still air and radiation give about 7.3 W/(m<sup>2</sup>·K), 1.11 W/K per cartridge, so the contents settle near 10 °C while the PCM melts and leave the 2 to 8 °C band after 2.4 h at 32 °C [D3] (4.0 h at 25 °C [D4]); the band holds indefinitely while the PCM melts only below about 19 °C ambient. A bare mill finish would give 1.9 h [D3], and a third dark cartridge 3.4 h [D5]. Under TCT-DDR-002 (O5) the user guidance is to lay produce directly on the cartridges, which is how ice packs are used; the model does not credit that contact, so this is a pessimistic bound, and R10 keeps its 8 h target until a TRL 4 measurement (on hold) shows the contact effect.

**R11 is met.** One H70 cartridge charged to 85 °C keeps 4 kg of food from 75 °C at 63 °C or more for 13.6 h with the decided dark finish and 7.2 h with a bare finish [D6]. The food alone would last 4.9 h [D7], so the cartridge adds at least 2.3 h. The hot case works because the H70 cartridge runs about 7 K above the food and radiation is stronger at 70 °C.

## ■ E. Recharge (R12)

A quasi-steady Stefan model adds the resistance of the growing layer of changed PCM ( $k$  0.2 W/(m·K), up to 22 mm from each broad face) to the air-side film.

*Table 4. Recharge time [E1, E2, E3].*

| Case                                  | Bare finish                                                                                       | Dark finish | R12 (8 h)     |
|---------------------------------------|---------------------------------------------------------------------------------------------------|-------------|---------------|
| C5 from 8 °C in a -18 °C freezer      | 5.6 h                                                                                             | 4.2 h       | Met           |
| C5 in a 2 °C refrigerator             | 51.1 h                                                                                            | 31.4 h      | Not practical |
| C25 from 28 °C in a -18 °C freezer    | 2.8 h                                                                                             | 2.1 h       | Met           |
| C25 in a 4 °C refrigerator            | 6.6 h                                                                                             | 4.6 h       | Met           |
| H70 from 25 °C in a fan oven at 85 °C | 6.1 h                                                                                             | 5.3 h       | Met           |
| H70 on a 60 W pad                     | 2.7 h if the heat reaches the PCM; up to 10.3 h if the melt must conduct upward from a 90 °C base |             | At risk       |

- The dark finish (O8) is the design case: C5 in a freezer 4.2 h, C25 in a refrigerator 4.6 h and H70 in an oven 5.3 h all meet the 8 h target [E1, E2].
- Charging C5 in a refrigerator, one option in the TRL 2 open questions, would take one to two days, so freezer charging with a conditioning step is the practical route (open item O3, no recommendation, still awaiting Amish).
- The TRL 2 figure of about 2.5 h on a 60 W pad assumed the heat is absorbed as fast as it is supplied. It needs 130 Wh [E3]; paraffin conducts poorly, and the pad must never push the base above the RT 70 HC limit of 90 °C, so the true time lies between 2.7 and 10.3 h and depends on convection in the melt. This is a question for a TRL 4 test.
- **R12 target correction.** The TRL 2 target allowed an oven "at 100 °C or less", which is above the 90 °C maximum operating temperature of RT 70 HC. TCT-REQ-001 v0.3 and later say 85 °C, never above 90 °C.
- The energy flow of one pad charge is 162 Wh in, 130 Wh stored between 25 and 80 °C, 88 Wh released above 63 °C and 42 Wh below it, with 32 Wh lost at the pad [E4] (Figure 2 of TCT-PRC-001).

## ■ F. Handle temperature (R14)

- **Thermal break.** Each bail lug sits on a 3 mm phenolic washer. The two washers conduct 0.032 W/K and the bail loses 0.069 W/K to the air, so a bail on a cartridge at 71 or 80 °C settles at 39.6 or 42.5 °C [F1], below the assumed 55 °C bare-metal threshold [F3].
- **Straight from the oven** the bail is at oven temperature, 85 °C. With a 10 min time constant it falls below 55 °C after about 12 min [F2]. The silicone grip is also at 85 °C at first, above the assumed 70 °C threshold for rubber.
- R14 is **met with the handling rule** decided in TCT-DDR-002 (O7): pad charging is preferred and keeps the bail near 42.5 °C, and the label tells users to take an H70 cartridge from an oven with oven gloves or to wait 12 min. Without the rule, a cartridge lifted straight out of an oven would not meet the burn threshold. The bail is bare aluminium on its thermal break and is not painted, so the dark finish does not change these figures. C5 and C25 never exceed 65 °C and meet R14.

## ■ G. Grade keying (R9)

Key tabs 30 mm wide sit at -43, 0, +43 and -61 mm (C5, C25, H70, W0), and each frame slot is 3 mm wider than its key on each side. Every pairing was checked: each frame accepts only its own grade [G1]. The W0 key overlaps the C5 slot but cannot pass it.

## ■ H. Cost and budget (R16)

- One cartridge costs \$76.00 against the \$50 target and one frame \$16.00 against \$20 [H1]. The dark finish (O8) adds about \$5 for primer and paint. The TRL 2 estimate of \$49 left out the fin bar and finish, priced the caps as simple plates and used a lower tube price; all figures remain indicative, not quoted.
- Decision D6 keeps the \$80 budget for one C5 cartridge and one frame. That build is \$92.00, \$12.00 over [H1]. Under TCT-DDR-002 (O6) supplier quotes come first and the budget is not raised until they are in; getting quotes is a purchasing step and waits with TRL 4, which is on hold. A set of one cartridge per grade and one frame would be about \$245 [H2]. PCM at \$20/kg instead of \$10/kg would add \$11 per cartridge [H3].
- The build uses plates cut and filed and bars bonded with epoxy, with no milling; the frame needs a hand brake, available at a makerspace.

## ■ I. Results against every requirement

*Table 5. Requirement status from this note [S1 and sections A to H].*

| ID  | Requirement                    | Value                                                                                                       | Target                                                | Status                               |
|-----|--------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------|
| R10 | Cold-chain hold time           | 2.4 h with the dark finish (1.9 h bare) at 32 °C; energy alone 9.7 h; produce contact not credited [D2, D3] | 8 h at 2 to 8 °C                                      | <b>Not met</b>                       |
| R16 | Low cost, garage-buildable     | \$76 per cartridge, \$16 per frame; no milling [H1]                                                         | \$50 and \$20                                         | <b>Not met</b><br>(cartridge)        |
| R12 | Recharge with common equipment | Dark finish: C5 freezer 4.2 h, C25 refrigerator 4.6 h, H70 oven 5.3 h; H70 pad 2.7 to 10.3 h [E1 to E3]     | 8 h or less                                           | <b>At risk</b> (pad)                 |
| R6  | Sealed for life                | Seal design reviewed; drop surge up to 5.0 bar flagged [C7]                                                 | No leak over 1,000 cycles and a 1 m drop              | Not verifiable at TRL 3              |
| R7  | Long life                      | Paraffins plausible; supplier cycling data not yet checked                                                  | 1,000 cycles, under 10 % loss                         | Not verifiable at TRL 3              |
| R17 | Open and documented            | Envelope, key and grade tables published (TCT-DWG-001, this note); fill recipe waits for TRL 4              | Published under CERN-OHL-S-2.0                        | Not verifiable at TRL 3 (partly met) |
| R1  | Standard grades                | RT 5 HC 5 to 6 °C, RT 25 HC 22 to 26 °C, RT 70 HC 69 to 71 °C (datasheets)                                  | C5 4 to 6, C25 22 to 27, H70 68 to 72 °C              | Met                                  |
| R3  | Stored energy                  | C5 65.4, C25 60.5, H70 72.9 Wh [B1]                                                                         | 60 Wh or more                                         | Met (C25 thin)                       |
| R4  | One-hand carry                 | 2.90 to 2.98 kg [A6]                                                                                        | 3.0 kg or less                                        | Met (thin)                           |
| R5  | Efficient use of mass          | 38.1 to 39.7 % PCM [A6]                                                                                     | 38 % or more (stock tube); 50 % for a later extrusion | Met (thin)                           |
| R8  | No pressure hazard             | 0.00 bar at fill temperature; +0.29 bar 20 K over the limit; walls 56 MPa under vacuum [C1 to C3]           | 0.5 bar or less, 10 % ullage                          | Met                                  |
| R11 | Hot-holding time               | 13.6 h with the dark finish (7.2 h bare) [D6]                                                               | 4 h at 63 °C or more                                  | Met                                  |
| R14 | Safe to handle when charged    | Bail 42.5 °C on pad charge; 85 °C for about 12 min after an oven, covered by the label rule [F1, F2]        | Below the 10 s burn threshold, with the H70 oven rule | Met with the handling rule           |
| R2  | Common envelope                | 323.4 x 152.4 x 63.5 mm [A1]                                                                                | GN 1/3, 65 mm                                         | Met                                  |
| R9  | Wrong grade cannot fit         | All pairings rejected [G1]                                                                                  | Keyed                                                 | Met by design                        |
| R13 | State visible                  | Sight window over a PCM vial                                                                                | Readable at a glance                                  | Met by design                        |
| R15 | Compatible materials           | Aluminium, FKM, epoxy rated 120 °C or more, paraffin only                                                   | No corrosion or swelling                              | Met by design<br>(paraffins)         |
| R18 | Recoverable                    | G 3/4 port drains the melt; fill named on label                                                             | Drain and reuse                                       | Met by design                        |

Counts: 2 not met, 1 at risk, 3 not verifiable at TRL 3, 12 met (8 by calculation or with a rule, 4 by design).

## ■ Checks against the TRL 2 figures

Table 6. TRL 2 claims checked.

| TRL 2 claim (TCT-PRC-001 v0.2)                        | This note                                                                 | Action                                               |
|-------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------|
| 322 x 152 x 61 mm                                     | 323.4 x 152.4 x 63.5 mm with the folding bail and stock 6.35 mm fin bar   | Precis and README updated                            |
| 1.62 L, about 1.1 kg PCM                              | 1.655 L; 1.10 to 1.18 kg                                                  | Updated                                              |
| C5 55, C25 61, H70 67 Wh                              | 65.4, 60.5, 72.9 Wh                                                       | Updated; R3 now met for C5                           |
| About 2.6 kg filled, 42 % PCM                         | 2.90 to 2.98 kg, 38 to 40 %                                               | Updated; R4 thin; R5 restated to 38 % by TCT-DDR-002 |
| Two C5 hold a 25 L cooler for about 8.1 h             | Energy lasts 9.7 h, but air-side coupling limits the hold to 1.9 to 2.4 h | R10 changed from at risk to not met                  |
| One H70 holds a GN carrier for about 6 h              | 7.2 h bare, 13.6 h with the decided dark finish                           | Updated                                              |
| C5 recharge 4 to 8 h in a freezer; H70 2.5 h on a pad | 5.6 h; 2.7 to 10.3 h                                                      | Updated; pad at risk                                 |
| Charge H70 at 100 °C or less                          | RT 70 HC limit is 90 °C                                                   | R12 target and safety text corrected                 |
| About \$49 per cartridge, \$15 per frame              | \$76 (with the dark finish) and \$16                                      | R16 changed from met to not met                      |